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Site-specific Mutagenesis of Human Apolipoprotein E

RECEPTOR BINDING ACTIVITY OF VARIANTS WITH SINGLE AMINO ACID SUBSTITUTIONS*

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Apolipoprotein (apo) E, an important protein involved in cholesterol transport in the plasma, binds with high specificity and high affinity to the apoB,E (low density lipoprotein) receptor. Several lines of evidence have indicated that key basic residues in the vicinity of residues 140–160 of apoE are important in mediating binding to the receptor. Furthermore, apoE variants exhibiting defective receptor binding are associated with the genetic lipid disorder type III hyperlipoproteinemia. To determine whether other basic amino acids in this region of apoE also affect receptor binding activity, site-specific mutagenesis of apoE in a bacterial expression system was undertaken. This system had been used successfully to produce apoE3 that was structurally and functionally equivalent to human plasma apoE3. Variants of apoE in which neutral amino acids were substituted for basic residues at positions 136, 140, 143, and 150 were produced. The variants all displayed defective binding; their activity ranged from 9 to 52% of normal (a range similar to that seen with naturally occurring variants of human apoE). In addition, to determine whether the conformation of this region is important for receptor binding, we designed variants in which proline was substituted for leucine 144 or alanine 152. Both variants were defective, exhibiting 13 and 27% of normal binding, respectively. In contrast, a double mutant in which arginine was substituted for serine 139 and alanine for leucine 149 displayed slightly enhanced receptor binding activity. These studies confirm that the middle of the apoE molecule is important in receptor binding and indicate that only certain amino acid substitutions in this region interfere with receptor binding activity.

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Apolipoprotein (apo)[1] E is associated with several classes of plasma lipoproteins (1, 2). Through its ability to interact with lipoprotein receptors, apoE is an important component of triglyceride and cholesterol metabolism (for review, see Ref. 1). Two types of polymorphism are found in apoE. One results from amino acid substitutions in the naturally occurring variants, and the second results from variable levels of sialylation (for review, see Refs. 2 and 3). Most of the known apoE variants have substitutions for basic amino acids and exhibit deficient binding to apoB,E(LDL) receptors (1, 2). The association of these variants with the lipoprotein disorder type III hyperlipoproteinemia has led to the suggestion that the defective binding contributes to this disorder (4, 5).

One region of apoE that is critical for the interaction with the apoB,E(LDL) receptor is in the vicinity of residues 140–160, as determined by studies of the primary structure of variants that bind defectively (2), the receptor binding activity of apoE fragments (6), and the inhibition of receptor binding by monoclonal antibodies (7). The secondary structure predicted for this region by the Chou-Fasman algorithm (8, 9) includes an α-helix (residues 131–150), a β-turn (residues 151–154), and a β-sheet (residues 155–163).

In the present study, we investigated the role of specific basic amino acids in this critical region and the effect of proline substitutions in this region on the receptor binding activity of apoE. Proline is predicted to disrupt the conformation at the site of the substitution. To address these questions, we produced apoE variants with single or double amino acid substitutions using an *Escherichia coli* expression system. This system had produced normal human apoE3 that possessed receptor binding activity equal to that of human plasma apoE3 (10). The apoE3 and apoE variants produced for this study were purified from the bacterial cells and complexed with the phospholipid dimyristoylphosphatidylcholine (DMPC). Previously, we had demonstrated that apoE·phospholipid complexes could be isolated and tested for binding to the apoB,E(LDL) receptor of fibroblasts (6, 11, 12).

MATERIALS AND METHODS

Production of Human Apolipoprotein E Variants—Human apoE variants with single amino acid substitutions at residues 136, 140, 143, 144, 150, and 152, and a double substitution at 139 and 149 were designed (Fig. 1*A*). To produce these variants, apoE cDNA cloned in M13 was mutagenized *in vitro* using oligonucleotide-directed site-specific mutagenesis according to Bauer *et al.* (14). The synthetic oligonucleotides used to introduce the mutations also contained a restriction enzyme recognition site as a convenient marker for the mutation (Fig. 1*B*). DNA sequencing (15) verified the various mutations. The mutated apoE cDNAs were expressed in *E. coli* using the system developed previously to produce apoE3 (10).

Purification and Characterization of Human Apolipoprotein E Produced in E. coli—The apoE variants were isolated from *E. coli* following sonication of the cells in the presence of dipalmitoylphosphatidylcholine (DPPC) and flotation of the phospholipid-bound protein by density gradient ultracentrifugation. Freeze-dried cell pellets were resuspended (50 mg of dry cells/ml) in a buffer (pH 7.4) containing DPPC (1.25 mg/ml), 150 mM NaCl, 20 mM Tris·HCl, 10 mM EDTA, 2 mM phenylmethylsulfonyl fluoride, and 0.01% Trasylol (Mobay Chemical). The cell suspension was sonicated five times for

[1] The abbreviations used are: apo, apolipoprotein; DMPC, dimyristoylphosphatidylcholine; DPPC, dipalmitoylphosphatidylcholine; LDL, low density lipoproteins.

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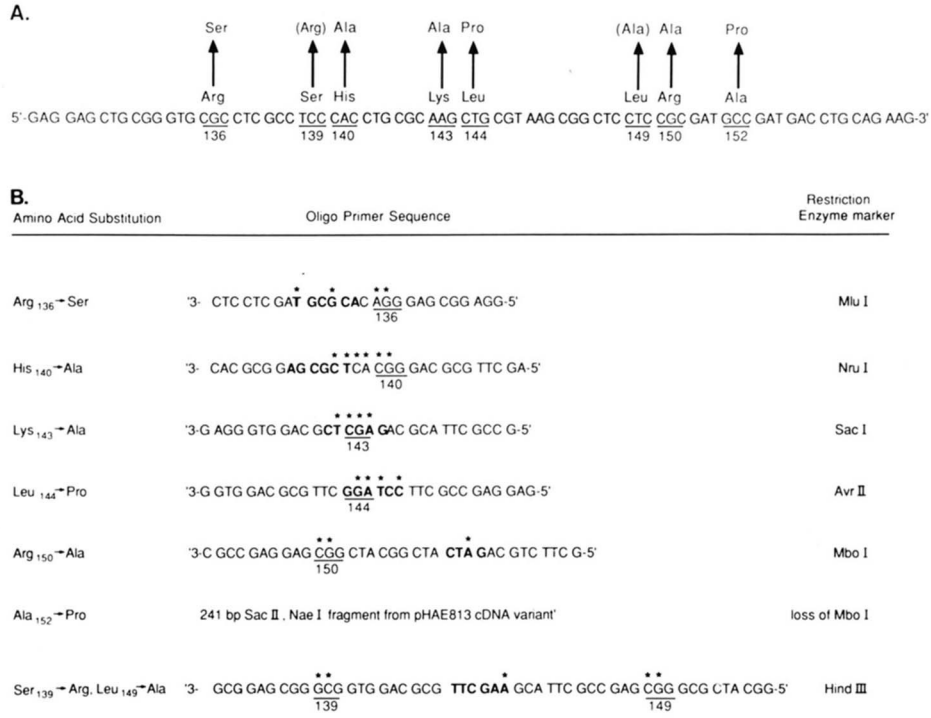


FIG. 1. *A*, human apoE3 cDNA sequence for amino acids 131–157. The genetically engineered substitutions are shown above the sequence (↑). The amino acid substitutions in *parentheses* indicate the double mutation. *B*, mutagenic oligonucleotide primers used for site-specific mutagenesis of human apoE cDNA. The primer sequences are complementary to that shown in *A*. *Asterisks* mark the mutations compared with the sequence in *A*. *Underlined* triplets with numbers mark the site of amino acid substitutions. The *boldface* sequence shows the restriction enzyme recognition site. For more details concerning the use of pHAE813 to construct the Ala^{152}→Pro variant, see Ref. 13.

2 min each. The suspension was allowed to warm up to 45 °C during the sonication and was cooled to 15 °C during the intervals. This temperature cycle was applied to facilitate the assembly of the apoE with the DPPC vesicles. The cell lysate was clarified by centrifugation at 10,000 × *g* for 20 min. The supernatant was adjusted to a density of 1.21 g/ml by adding solid KBr and centrifuged for 12 h at 16 °C in a 60 Ti rotor (Beckman) at 60,000 rpm. The turbid band that floated at the top was collected, dialyzed against 25 mM NH_4HCO_3, lyophilized, and delipidated using the chloroform/methanol extraction procedure developed for plasma lipoproteins (16). The delipidated protein was dissolved in 6 M guanidine·HCl containing 150 mM NaCl, 20 mM Tris·HCl, 10 mM EDTA, and 10 mM dithiothreitol (pH 7.6) and fractionated by gel filtration on Sephacryl S-300 (Pharmacia LKB Biotechnology Inc.) and by affinity chromatography on heparin-Sepharose (16). The purified apoE was dissolved in 100 mM NH_4HCO_3, and stored at −20 °C. Gel electrophoresis (17) and isoelectric focusing (18) were performed as described.

Preparation and Analysis of Apolipoprotein E·DMPC Complexes—Plasma apoE3 and bacterially produced apoE were combined with DMPC vesicles, floated by ultracentrifugation, obtained from the gradient at d = 1.08–1.10 g/ml, and analyzed as previously described (11, 12). Protein was determined according to the method of Lowry *et al.* (19). The apoE·phospholipid discs were characterized following negative staining by electron microscopy (20), and the size of the discs was determined from the photo images using a calibrated magnifying glass. The receptor binding activity of the apoE·DMPC complexes was determined in a competitive assay with ^{125}I-labeled human LDL, using human primary fibroblasts in culture (21).

RESULTS AND DISCUSSION

Isolation and Characterization of the Bacterially Produced Apolipoprotein E3 and Apolipoprotein E Variants—Purification of the *E. coli*-produced apoE was monitored by acrylamide gel electrophoresis. As shown in Fig. 2 for a typical isolation, a band in the region of apoE was not apparent in the total *E. coli* extract (*lane 1*); however, it could be seen in the supernatant of the cells lysed in the presence of DPPC (*arrow*, *lane 2*). Partial purification was possible following ultracentrifugal flotation of the DPPC-complexed proteins from the lysate (*lane 3*). This apoE-enriched fraction was then delipidated, and the purification was completed by gel filtration (Sephacryl S-300) and heparin-Sepharose chromatography. The

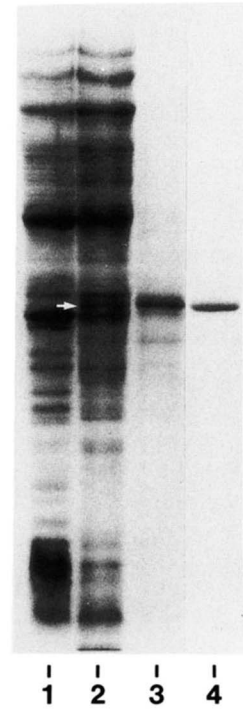


FIG. 2. Coomassie Blue-stained acrylamide gels demonstrating the steps in the purification of apoE variant Lys^{143}→Ala: *lane 1*, total cell extract; *lane 2*, supernatant of cells lysed in the presence of DPPC, with *arrow* indicating band in the region of apoE; *lane 3*, d < 1.21 g/ml turbid band that floated at the top of the density gradient after ultracentrifugation; *lane 4*, purified apoE variant Lys^{143}→Ala following Sephacryl S-300 gel filtration and heparin-Sepharose chromatography.

apoE obtained by this protocol migrated as a single band, *lane 4*, that comigrated with plasma apoE3.

The isoelectric focusing positions of the apoE variants relative to apoE3 reflected the charge alterations expected for the various amino acid substitutions (Fig. 3). All the bacte-

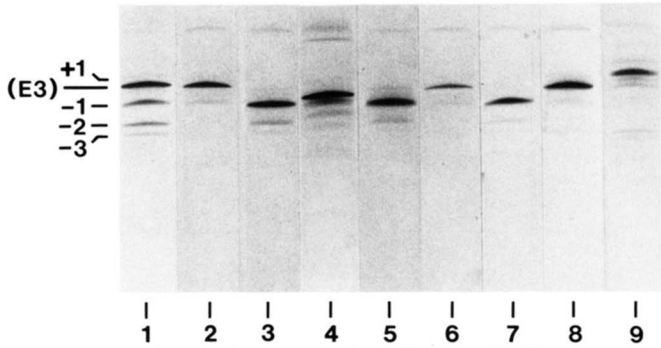


FIG. 3. **Isoelectric focusing of purified apoE.** Plasma apoE3 (*lane 1*) and its sialylated isoforms are designated *E3* and *−1*, *−2*, and *−3*, respectively. *Lanes 2–9*, bacterially produced apoE variants: *lane 2*, normal apoE3; *lane 3*, Arg^{136}→Ser; *lane 4*, His^{140}→Ala; *lane 5*, Lys^{143}→Ala; *lane 6*, Leu^{144}→Pro; *lane 7*, Arg^{150}→Ala; *lane 8*, Ala^{152}→Pro; and *lane 9*, Ser^{139}→Arg, Leu^{149}→Ala. *+1*, positive charge, gained by double mutation (*lane 9*).

rially produced apoE variants focused primarily in a single band, indicating high purity. Characteristically, the plasma-derived human apoE3 focused in a major band (Fig. 3, *lane 1*), representing the nonsialylated form, and in three minor bands, representing the more acidic sialylated isoforms (designated as *−1*, *−2*, and *−3* in Fig. 3) (2, 3). The variants with a substitution of a neutral for a basic amino acid at positions 136, 143, and 150 (*lanes 3*, *5*, and *7*, respectively) cofocused with the −1 isoform of the plasma apoE3, indicating loss of a single positive charge in these variants. The double mutation involving substitutions at positions 139 and 149 gained a positive charge and focused at the +1 position (*lane 9*), as would be predicted. The variants involving proline substitutions (*lanes 6* and *8*) produced no charge alterations and cofocused with the nonsialylated major isoform of plasma apoE3. The variant in which alanine was substituted for histidine at residue 140 focused in a position between plasma apoE3 and the −1 position, suggesting that the histidine at position 140 in normal apoE3 is only partially protonated.

Isolation and Characterization of Apolipoprotein E·DMPC Complexes—Apolipoprotein E requires the presence of phospholipid to display receptor binding activity (11). The procedure used to form apoE·phospholipid discs (6, 11, 12) results in apoE3 complexes whose affinity for the apoB,E(LDL) receptor is equivalent to that of naturally occurring canine apoE-containing high density lipoproteins (HDL), *i.e.* apoE HDL_c (11, 12). Furthermore, apoE·DMPC complexes are useful in comparing naturally occurring apoE variants that are associated with type III hyperlipoproteinemia (22, 23).

The purified apoE variants produced by site-specific mutagenesis were combined with DMPC and floated by density gradient ultracentrifugation to $d = 1.08–1.10$ g/ml, consistent with previous results. As shown in Table I, the bacterially produced apoE tended to form apoE·DMPC discs that were slightly larger than the plasma apoE3 discs. However, all of the apoE variants, except the proline variants, formed discs of similar size. The discs formed with the proline variants were larger and somewhat more heterogeneous.

Receptor Binding Activity of the Site-specific Variants—The receptor binding activity of the various apoE·DMPC complexes was determined in a competitive assay using ^{125}I-labeled human LDL and cultured fibroblasts. As previously demonstrated, the activity of the bacterially produced apoE3 was essentially identical to that of plasma-derived apoE3 (10), and, therefore, all binding activities were normalized to that activity (Table II). The neutral substitutions for basic residues presented in this study are located in a region postulated to

TABLE I

Electron microscopic characterization of the human apolipoprotein E·phospholipid complexes

ApoE	Disc diameter[a]
	nm ± S.D.
Plasma apoE3	14.4 ± 2.6 ($n = 3$)
Bacterially produced variants	
ApoE3	17.2 ± 4.2 ($n = 3$)
Arg^{136}→Ser	16.4 ± 3.1 ($n = 3$)
His^{140}→Ala	17.5 ± 4.1 ($n = 3$)
Lys^{143}→Ala	16.8 ± 4.0 ($n = 3$)
Arg^{150}→Ala	16.7 ± 4.0 ($n = 3$)
Leu^{144}→Pro	23.8 ± 5.3 ($n = 3$)
Ala^{152}→Pro	22.9 ± 5.5 ($n = 3$)
Ser^{139}→Arg, Leu^{149}→Ala	18.7 ± 4.7 ($n = 2$)

[a] More than 500 particles were measured for each experiment. n = number of experiments.

TABLE II

Receptor binding activity of bacterially produced variants of human apolipoprotein E

Bacterially produced apoE variants	50% displacement of ^{125}I-LDL[a]	Percent of normal receptor binding activity[b]
	ng protein/ml ± S.E.[c]	
ApoE3	43 ± 4 ($n = 4$)	100
Arg^{136}→Ser	105 ± 25 ($n = 3$)	40
His^{140}→Ala	82 ± 14 ($n = 3$)	52
Lys^{143}→Ala	480 ± 165 ($n = 4$)	9
Arg^{150}→Ala	174 ± 56 ($n = 3$)	24
Leu^{144}→Pro	336 ± 205 ($n = 4$)	13
Ala^{152}→Pro	159 ± 32 ($n = 4$)	27
Ser^{139}→Arg, Leu^{149}→Ala	28 ± 3 ($n = 3$)	154

[a] The concentration of apoE required to displace 50% of the ^{125}I-LDL from the apo-B,E(LDL) receptors on human fibroblasts.

[b] Expressed as a percentage of the activity of bacterially produced apoE3 and calculated from the concentration required to displace 50% of the ^{125}I-LDL from the cells.

[c] The standard error (S.E.) of the geometric mean was calculated by multiplying the value of the geometric mean by the S.E. of the averaged logarithmic transform (24).

be an α-helical structure. Alanine was selected to minimize the effects of bulky side chains and also to preserve the postulated α-helical structure of the region. Serine was selected to substitute for arginine at residue 136 because a naturally occurring variant with this substitution was reported in a compound heterozygous patient with type III hyperlipoproteinemia (25). Substitution of neutral amino acids for basic amino acids at positions 136, 140, 143, and 150 reduced binding activity by various amounts (Table II).

These results suggest that the basic residues 136, 140, 143, and 150 are involved in binding and complement previous studies with naturally occurring variants that indicated the involvement of basic residues 142, 145, 146, and 158 (for review, see Ref. 2). Except for the variant with a substitution at residue 158, most of the variants demonstrate a moderate decrease in binding activity when single basic residues are mutated. The fact that no single substitution results in complete abolition of binding activity suggests that there are multiple interactions between apoE and the receptor and that the basic residues may cooperate in the binding function.

The ligand binding portion of the apoB,E(LDL) receptor possesses seven repeats highly enriched in acidic amino acids (for review, see Ref. 26). These repeats have been suggested to be possible sites of interaction with apoE, as well as with apoB. It is likely that ionic interactions play a major role in the binding of apoE to the apoB,E(LDL) receptor. Support for the ionic nature of the interaction between the ligand and the receptor is provided also by experiments demonstrating

that polyanionic agents such as polyphosphates and heparin inhibit this binding (for review, see Ref. 21). Therefore, the binding between apoE and the apoB,E(LDL) receptor may involve the direct ionic interaction of an array of oppositely charged residues (arginyl, lysyl, and possibly histidyl residues of apoE with aspartyl and glutamyl residues of the receptor). Although the neutral substitutions (alanine or serine) were selected to minimize the possibility of conformational disruptions, we cannot exclude the possibility that one or more of the substitutions decrease receptor binding by conformational disruption of the putative receptor binding domain. It is possible, for example, that interactions of this region with other parts of the apoE polypeptide have been disrupted by destruction of an important charge-charge interaction or the creation of a new hydrophobic interaction. To determine the mechanism by which these substitutions affect the receptor binding of apoE, we are analyzing in detail the structure of the variants. Furthermore, the variable effects of the single substitutions on receptor binding may be significant; thus, some basic residues may be more critical for binding than others. To address this question and the suggestion that multiple interactions between apoE and the receptor are required for binding, we are preparing additional variants. For example, we will assess the relative contribution to receptor binding of the basic residue doublet at positions 142, 143 *versus* that of the basic residue triplet at 145, 146, 147.

In contrast to the moderate effects on binding of substitutions at positions 136, 140, 142, 143, 145, 146, or 150, the naturally occurring variant with cysteine substituted for arginine at residue 158 demonstrates a substantial decrease in binding (to 1–2% of normal). However, it has been suggested that the effect of this substitution is primarily a conformational one that disrupts the integrity of the receptor binding domain (27). Therefore, the involvement of arginine at position 158, which lies outside of the putative α-helix, may be of a different character than the basic residues within the proposed α-helical segment in that arginine 158 may not be a "direct contact" residue.

The importance of the conformation for receptor binding activity was probed further by analyzing the receptor binding activity of apoE variants in which proline was substituted at two key positions. The substitution of proline at position 144 or 152 was predicted to affect the conformation in the vicinity of the substitution. Position 144 is located in the center of the cluster of basic residues involved in receptor binding. The region was postulated by the Chou-Fasman algorithm to consist of an α-helix in which the basic residues are clustered on one of its faces (27). Substituting proline at position 144 was expected to break the helical structure and disrupt the continuity of the positively charged cluster. Indeed, the receptor binding activity of this variant was defective (Table II). This result is consistent with the view that the multiple basic residues in this region may require a particular orientation for optimal receptor binding activity.

The apoE variant with proline at position 152 had been deduced from a cDNA sequence cloned from a human liver cDNA library (13). The donor was not available for further analysis, but indications of hyperlipidemia in family members of the donor increased our interest in studying the receptor binding characteristics of an apoE variant with this substitution. Residues 151–154 in apoE were postulated to form a β-turn (27). Thus, position 152 would be located in a "hinge-like element." Because proline at this position would enhance the probability of β-turn formation, its impact on the conformation of apoE was less predictable than at 144. However, the moderate effect of this substitution on receptor binding activity (Table II) suggests that some alteration affecting the orientation of the domains neighboring this "hinge" had occurred. It has been demonstrated that substituting proline for threonine at position 51 in tyrosyl tRNA synthetase has a purely local effect on the structure of the enzyme (28).

The results with the proline variants demonstrate that receptor binding can be reduced without altering positively charged residues. The slight alteration in the structure of the DMPC complexes of these variants supports the possibility that the decreased receptor binding is probably due to conformational changes. However, explicit understanding of the molecular mechanism by which the proline substitutions affect the receptor binding of apoE requires complementary studies involving detailed structural analysis of the variants.

Finally, the apoE variant with substitutions at positions 139 and 149 displays slightly higher binding activity than apoE3 (Table II). In the context of the present study, this variant demonstrates that not all substitutions in this region reduce the receptor binding activity of the molecule.

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